Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in optical/materials characterization, thin-film process development, DOE/SPC, and failure analysis. Builds reliable measurement workflows using spectroscopy, ellipsometry, thermal analysis, electrochemistry, and structured data to connect material signatures with process improvements.

CORE COMPETENCIES

Materials Characterization | Solar/PV Metrology Ramp | Optical Spectroscopy | Ellipsometry | Root-Cause Failure Analysis | DOE / SPC | SOPs and Calibration | Process-Quality Feedback

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided process-development experiments for bio-based polymer systems, defining material-quality readouts, process windows, reproducibility criteria, and scale-up specifications.
- Translate FT-IR, thermal-analysis, spectroscopy, and process data into SOP-ready procedures and technical recommendations for repeatable execution.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Assessed reliability and degradation failure modes across 15+ functional material variants using spectroscopy, electrochemistry, thermal analysis, NMR/GPC, and MS-informed characterization.
- Built process-property datasets connecting purity, stability, optical response, and performance to experimental conditions, anomalous results, and material-design recommendations.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded molecular thin-film coating process development, optimizing CVD/surface-preparation conditions to improve reproducibility, morphology, interface quality, and optical/material performance.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Developed polymer thin-film deposition under flow using DOE strategies and in-situ QCM-D monitoring to quantify growth kinetics and coating reproducibility.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, safety, and equipment-tracking systems that improved instrument uptime by 30% and strengthened data quality.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Electrochromic Material Reliability Study
Chemistry of Materials 2025

Mapped optical response, degradation, and performance failure modes across functional material variants with orthogonal characterization.

UV-Vis-NIR, FT-IR, CV, TGA/DSC, NMR, GPC, MS

AFOSR Defense Thin-Film Coatings
DoD Funded

Built process-property understanding for thin-film coatings by connecting CVD variables, surface preparation, morphology, and optical performance.

CVD, QCM-D, ellipsometry, surface prep, DOE

Bio-Based Polymer Scale-Up Process
Current

Defined process windows, measurement readouts, reproducibility criteria, and SOP-ready quality specifications for polymer scale-up.

DOE, SPC, FT-IR, thermal analysis, technical reports

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Characterization & Metrology: UV-Vis-NIR, FT-IR, ellipsometry, spectroelectrochemistry, QCM-D, TGA, DSC, NMR, GPC/SEC, LC-MS, GC-MS, optical/materials data interpretation
Reliability & Failure Analysis: Failure mode identification, root cause analysis, degradation analysis, stress-test design, DOE, 5-why/fishbone-adjacent troubleshooting
Process & Quality Systems: SPC, process windows, SOPs, calibration systems, reproducibility studies, measurement acceptance criteria, technical reports
Materials: Thin films, CVD coatings, polymers, organic/organometallic materials, surface functionalization, structure-property-processing relationships
Data & Tools: Excel, Python/basic, JMP/basic, structured experimental records, dashboards/visualization concepts, publication-quality communication

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)